



Jędrzej Cader

Mechanical Engineer, Coach, Researcher

Professional Experience

Researcher

03.2025-11.2025

NASA JPL – Jet Propulsion Laboratory

- Worked as a Visiting Student Researcher in the Robotics Group at JPL, developing the signal processing pipeline for the Lunar Crater Radio Telescope (LCRT),
- Developed an orbital simulations toolkit and an antenna/foreground convolved signal extraction pipeline,
- **Skills:** Matlab, Git, Python, software collaboration, adaptability,

Head Coach & Technical Advisor

08.2024-present

ARIS – Swiss Academic Spaceflight Initiative

- Leading the coaching and technical advisor team for flagship rocketry project HERMES, building a bi-liquid rocket at ARIS
- **Skills:** leadership, project management, bi-liquid rocketry,

Mechanical Engineer

11.2023 -10.2024

ARIS – Swiss Academic Spaceflight Initiative

- Worked as a propellant supply engineer on a liquid rocket engine test stand project – Project HELIOS,
- Co-designed and implemented a throttling control mechanism and cryogenic liquid oxygen valve systems,
- Successfully led and executed rocket engine test campaigns, demonstrating proficiency in team coordination, fluid dynamics,

Teaching Assistant (TA)

12.2021-09.2022

ETH Zürich

07.2023-04.2024

- Successfully led and mentored six teams of five students in the Innovation Project at ETH Zurich, providing technical, strategic, and social guidance,
- Fostered and imparted expertise in robot and mechanical systems, project planning, as well as agile and innovative product development,

Engineering + Workshop Intern

07.2023-09.2023

KMB Grupa, SOMET

- Acquired expertise in designing and manufacturing parts for manual and CNC manufacturing – CAD and CAM.
- Hands on experience in stainless-steel workshop manufacturing stainless steel kitchen furniture,

Education

Master in Mechanical Engineering,

2024-present

ETH Zürich

- Specialising in Robotics, Design and Aerospace subjects,
- **Research Project:** Wake-Based Analysis and Distributed Control of a Drone Swarm,
- **Semester Thesis:** Testbed Development for Aerospace Micropumps. Laying foundations for aerospace pump research at ARIS and the ETH.
- **Skills:** CAD, robotics systems, aerospace engineering, rocketry, research, Python.

Bachelor in Mechanical Engineering

2020-2024

ETH Zürich

- Focus on Robotics, Design, and Control,
- **Bachelor's Thesis:** on Additively Manufactured Cryogenic Ball Valves for Rocket Engines,
- **Skills:** Computational Fluid Dynamics (CFD), robotics, control systems, Matlab.

ABOUT ME

I am an ambitious Mechanical Engineer studying at the ETH Zürich, with a passion for **aerospace, research and innovation**. Complementing my studies, I work as an engineer and advisor, developing and testing **bi-liquid rockets and rocket engines**, pushing myself beyond just studying. As a **Teaching Assistant** at ETH, I have honed my ability to communicate complex technical concepts and work in a team. **Working at NASA**, I challenged myself and successfully contributed to leading global space research.

Fluent in four languages with experience living in three countries, I bring adaptability and open-mindedness to every team. My leadership, cooperation and organizational abilities were honed through various roles within the European Youth Parliament (EYP), including President and Chair, where I successfully led and organized events for over 100 participants. Additionally, my eight years as a tutor have enhanced my problem solving, teamwork and communication. Overall, I bring a useful combination of **technical expertise, passion and collaboration** to each endeavour.

As a former professional swimmer, I am highly driven and results oriented. My aspiration is to **push forward the frontier** of space technology and research, to help humanity take to the stars.

KEY SKILLS

- **CAD (e.g. Siemens NX, Fusion360, Solidworks),**
- **CAM (e.g. Mastercam, Fusion360),**
- **Languages:** English, Polish, German, French,
- **Software:** Python, Simulink, Matlab, Git
- **FEA:** ANSYS CFD, ANSYS Mechanical,
- Management (people and project), project organisation,
- Teamwork, interpersonal relations, collaboration

CONTACT

 jcader@ethz.ch

 www.linkedin.com/in/jedrzej_cader/

LANGUAGES

- English: [Progress bar] Native, fluent
- Polish: [Progress bar] Native, fluent
- German: [Progress bar] Full casual and professional fluency
- French: [Progress bar] B2/C1: good professional fluency

KEY SKILLS

- CAD (e.g. Siemens NX, Fusion360, Solidworks),
- CAM (e.g. Mastercam, Fusion360)
- CFD (ANSYS)
- Software languages: Python, Simulink
- Leadership + Management, project organisation
- Interpersonal relations, people management

HOBBIES

-  Swimming
-  Travelling
-  Teaching, mentoring
-  Public speaking
-  Learning new skills
-  Puzzle solving; chess, Rubik's cubes

CONTACT

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 Switzerland

Education

- Hull's School English College** **2016-2020**
 - Graduated with straight A* in Mathematics, Physics, Economics and French,
 - Founder of Debate Society,
 - Finalist of Swiss Mathematics and Physics Olympiads.

Volunteering, Projects, Extracurriculars

- Project Manager, Volunteer** **2018-present**
European Youth Parliament:
Organised and managed large-scale debating events aimed at bringing the European youth together in the realm of sociopolitical issue,
 - Presided** a 100+ people debating event (session) in Sassari, Italy, 2023. Oversaw the performance of participants, gave lectures on leadership and chairing,
 - Led** seven different 10+ people committees through teambuilding and debating and drafting proposals,
 - Organised and presided a smaller scale, 20 people events in Tessin, Switzerland,
 - Skills and competences:** leadership, public speaking, **people management, teamwork, collaboration, creativity, problem solving.**

- Co-founder, External Relations Manager, Polish Academic Network Abroad (POLANA):** **2023-present**
 - Co-founded** the biggest Polish Student Association in Zürich, (polana.ch)
 - Responsible for outreach, partner acquisition, events,
 - Successfully contributed to building an organisation of 200+ members,
 - Contacted sponsors, organised events for 100 people,

- Academic Tutor** **2017-present**
 - Successfully mentored and taught high school and university students in Mathematics, Physics and Economics, Consistently achieving exceptional results and exceeding academic expectations.

- Mentor** **2022-present**
Mentorship ETH Zurich:
 - Serving as a mentor in the ETH Zürich's Mentorship Programme, providing guidance and support to first-year students,
 - Developed strong communication, teaching and teamwork skills through this role.

- Engineer,** **2021-2022**
Innovation Project ETH Zurich:
 - Built a desktop snack dispenser robot as part of the ETH Innovation Project, in a team of 4 people, applying an agile product development framework.

- Professional Swimmer:** **2012-2021**
Polish Junior National Team
 - Represented Poland as part of the Polish Junior National team,
 - Achieved National championships in 200m butterfly, 4x100m freestyle and 4x100m medley,
 - Developed exceptional grit, perseverance, ambition, competitive drive and a goal-oriented mindset through high level athletic training.